

Wall-Charge Transport under Flux: Exact Benchmark, Static Response, and Monitored-Circuit Ramps

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¹*Class-A fermionic adaptive-circuit project*

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We collect the present numerical evidence for wall-resolved charge transport under a threaded boundary flux and specify the next monitored-circuit test. For the exact flattened Chern-insulator Hamiltonian, an independently filled instantaneous ground state closes after one flux quantum, whereas overlap continuation through the wall crossing transfers almost one fermion between the two interfaces. At 20×40 , the continued soft-wall response is $q_x = 0.98397186$ for increasing flux and -0.99999998 for decreasing flux; the hard-wall values are 0.98080970 and -1.00000000 . A completed 16×20 monitored-circuit experiment instead imposed one fixed Born record in independent static-twist replays. It displays an order-one, direction-sensitive conditional response, with maximum $|q_x| = 0.62807$ for the soft wall and 0.43335 for the hard wall, but closes at 2π and is not a dynamical pump. A completed ten-trajectory online pilot subsequently branched each cycle-40 burn-in state into independently Born-sampled clockwise and counterclockwise ramps. We now preregister a paired conditional test froze one zero-flux continuation record from each verified burn-in state and replayed it continuously along both flux orientations for $M = 16, 32, 64$ cycles. Its direction-odd endpoint decreases with M , rather than settling to a nonzero adiabatic value. We therefore preregister a fixed-flux quench: the same burn-in frame and 64-cycle record are replayed with every OW measurement held at one of four signed twists, including $2\pi - 10^{-7}$. This directly tests the transient response of a conditioned measurement channel without tangent-mode tracking. A separate state-derived Hamiltonian experiment on 50 saved 20×24 endpoints has now completed. Finite-time RK4 evolution at $T = 10^4$ gives raw endpoint means $+0.71569$ (CCW) and -0.71422 (CW) for the soft wall, and $+0.53622$ and -0.53624 for the hard wall. Its planned 24×24 width repeat was stopped with zero completed paths and 28 exact rolling checkpoints preserved at interval 104. We now preregister an adaptive Kato experiment at both widths. It removes physical ramp time, defines the spectral branch by previous-projector overlap, and reports raw CW and CCW transfer separately; no Kato result is claimed here. A later wall-diabatized spectral calculation is now complete for 1600 monitored endpoint states. Across its 3200 raw directional paths, 98.3% have the correct orientation and $|q_x| > 0.9$, while 97.5% lie within $0.95 \leq |q_x| \leq 1.05$. This final calculation is a spectral-flow property of flattened endpoint projectors, not a real-time monitored pump.

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I. PURPOSE AND CLAIM STRUCTURE

Threading a flux through a cylinder supplies a direct test of chiral boundary transport in a Chern insulator [1, 2]. In an equilibrium Hamiltonian, the relevant adiabatic object is a continuously transported occupied subspace. In a monitored adaptive circuit, the state is instead changed by Born-conditioned measurements and fermionic feedback, so a static family, an online ramp, and modular-time packet propagation answer different questions. This working note keeps those distinctions explicit.

The evidence is organized in complementary layers. First, the exact flattened Hamiltonian provides a controlled equilibrium benchmark. Second, the completed frozen-record calculation measures the circuit's conditional static response. Third, the completed $S = 10$ experiment makes the twist time dependent inside one physical monitored trajectory while independently sampling its directional branches. Fourth, the continuous frozen-record ramp holds the complete measurement record fixed while changing only the flux path. Fifth, the fixed-flux quench defined below removes the ramp and tests the transient conditioned channel at four signed twists. Sixth, a completed finite-difference unitary experiment evolves saved circuit endpoints under their state-derived parent Hamiltonians. Seventh, the newly preregistered Kato calculation transports the overlap-selected spectral projector directly in flux. Eighth, the completed wall-diabatized width sweep isolates the physical wall doublet from ordinary spectator states and preserves its entering wall label through the finite-size crossing. These constructions answer different physical questions and their outputs are not silently pooled.

II. COMMON GEOMETRY AND CHARGE OBSERVABLE

The physical cylinder has two orbitals per unit cell, is open in x , and is periodic in y . A topological slab with parent-Hamiltonian parameter $\alpha_1 = 1$ is embedded in a trivial exterior with $\alpha_2 = 30$. We use the overcomplete Wannier (OW) basis with trial family X and $n_{\text{shell}} = 1$. Distinct OW number stabilizers need not commute, and the OW modes form a localized overcomplete frame rather

than an orthonormal Wannier basis [3].

Let G be the physical single-particle correlation matrix,

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j), \quad G = \frac{\mathbf{1} + \Gamma}{2}, \quad (1)$$

where Γ is the centered covariance used by the numerical engine. The cylinder is divided into the disjoint half-system regions R_L and R_R . For a path beginning at flux ϕ_0 , define

$$\begin{aligned} \Delta N_a(\phi) &= \text{tr}[R_a G(\phi)] - \text{tr}[R_a G(\phi_0)], \quad a = L, R, \quad (2) \\ q_x(\phi) &= \frac{\Delta N_R(\phi) - \Delta N_L(\phi)}{2}. \end{aligned} \quad (3)$$

If total charge is conserved, $\Delta N_L + \Delta N_R = 0$ and therefore $q_x = \Delta N_R = -\Delta N_L$. The scalar q_x is a signed left-right redistribution; its interpretation depends on how the family of states $G(\phi)$ is constructed.

III. EXACT FLATTENED-HAMILTONIAN BENCHMARK

A. Instantaneous and continued projectors

The exact benchmark uses a 20×40 cylinder with topological slab $x \in [5, 15]$. At each flux, the flattened one-particle Hamiltonian is diagonalized in complex128. Two occupied-state prescriptions are compared. The *instantaneous* projector fills the lowest fixed number of modes independently at every ϕ . Because $\hat{H}(2\pi)$ is related to $\hat{H}(0)$ by a large gauge transformation, this finite-system ground-state projector closes after one loop. The *continued* projector instead matches the occupied frame between neighboring fluxes by maximum overlap and therefore follows the wall branch through its crossing. This is the finite-size realization of adiabatic spectral flow.

The physical-wall crossing at zero flux is exponentially small. A signed offset $\varepsilon = 10^{-7}$ selects an unambiguous side of that crossing:

$$\phi_j^{(\sigma)} = -\sigma\varepsilon + \sigma \frac{2\pi j}{16}, \quad \sigma = \pm 1. \quad (4)$$

This offset is a branch-selection convention, not a mass or broadening. The quantization-favoring origin lies on

the side opposite the direction of winding. Reversing the origin changes the small finite-size deficit but not the direction of transport.

B. Equilibrium results

Figure 1 shows that the instantaneous family redistributes charge during the loop but returns to its starting projector. On the 257-point grid, its endpoint $|q_x|$ is at most 1.94×10^{-8} for the soft wall and 4.13×10^{-9} for the hard wall. Closure is therefore a property of independently refilling the ground state, not evidence against the intervening wall crossing.

The continued family instead leaves one particle-hole rearrangement at the endpoint. For the soft wall, increasing flux gives $(\Delta Q_L, \Delta Q_R, q_x) = (-0.98397186, 0.98397186, 0.98397186)$, while decreasing flux gives approximately $(1, -1, -1)$. The hard-wall values are $(-0.98080970, 0.98080970, 0.98080970)$ and $(1, -1, -1)$. The transferred density is localized at the two physical interfaces, and reversing the flux reverses the transfer. The residual $|\Delta Q_L + \Delta Q_R|$ is at most 1.14×10^{-13} .

The finite-size and regulator checks use $N_y = 16, 24, 40, 65$ flux points, $\phi_0 = \pm 10^{-7}$, and both soft and hard wall constructions. With the quantization-favoring signed origin, the forward magnitudes differ from unity by at most 2.4×10^{-9} over this size set. Starting from the opposite side gives finite-size deficits ranging from 0.00262 to 0.01919; the deficit grows with N_y because the chosen initial branch has a small opposite-wall component. A true reverse traversal initialized from the continued endpoint returns exactly to the initial transported frame within a projector Frobenius error per dimension below 4.2×10^{-18} . The flattened Hamiltonian is Hermitian to stored precision, its off-block residual is below 1.3×10^{-15} , and its large-gauge closure error is below 7.3×10^{-15} .

IV. COMPLETED STATIC MONITORED-CIRCUIT RESPONSE

A. Protocol

The monitored calculation uses a 16×20 cylinder with topological slab $x \in [4, 12]$. One soft-wall and one hard-wall trajectory were sampled for 40 cycles from a pure half-filled random state, with random-serial ordering, perfect correction, no postselection, and complex128 arithmetic. Each trajectory's complete zero-twist Born record was then imposed in 68 independent static reruns: 17 twists in each direction for each wall. The OW functions and initial state were rebuilt in a common uniform twist gauge for each rerun. Neighboring flux points were not dynamically connected.

The signed origin followed Eq. (4). Because each static rerun imposed the same full record, the total feedback

charge was fixed within an arm: -2 for the soft record and -4 for the hard record. The plotted ΔN_L , ΔN_R , and q_x are direct differences from the direction-specific origin. No source-subtracted curve is used in this note.

B. Static-response results

The two records show substantial wall redistribution. The maximum response is

$$\max_{\phi} |q_x| = \begin{cases} 0.62806884, & \text{soft wall,} \\ 0.43334984, & \text{hard wall.} \end{cases} \quad (5)$$

The left and right curves are equal and opposite to numerical precision, with maximum regional-balance residual 8.53×10^{-14} . Global charge continuity holds to 1.14×10^{-13} . After a full flux quantum, all four wall/direction paths close: the large-gauge covariance mismatch per dimension lies between 1.54×10^{-16} and 2.65×10^{-16} .

The correct present conclusion is consequently narrow but positive: the conditioned monitored circuit has an order-one, handed wall response in both wall implementations. The experiment does not demonstrate transported charge over a dynamical cycle, because every ϕ is a separate static replay and the state closes at 2π . The use of only one record per wall is an additional reason not to infer an ensemble-level pump coefficient.

V. PREREGISTERED ONLINE-RAMP PILOT

A. Scientific contract

The online pilot retains $N_x \times N_y = 16 \times 20$, $n_{\text{shell}} = 1$, $(\alpha_1, \alpha_2) = (1, 30)$, pure half filling, perfect correction, no postselection, and complex128. It changes the site order to raster- y and samples ten independent parent trajectories for each of the soft and hard walls. Each parent evolves for $T_{\text{burn}} = 2N_y = 40$ cycles at exactly zero flux. Its exact occupied frame after cycle 40 is then copied into two branches. The branches share that physical state but use independently derived random streams for their subsequent Born outcomes.

For direction $\sigma = +1$ (counterclockwise) or -1 (clockwise), the online schedule is

$$\phi_0 = 0, \quad \phi_j^{(\sigma)} = \sigma \frac{2\pi j}{16}, \quad j = 1, \dots, 16. \quad (6)$$

The OW projectors are rebuilt in the uniform gauge immediately before the first site update of each ramp cycle. There is no 10^{-7} offset: the online trajectory starts from the actual zero-flux burn-in state, rather than selecting a spectral branch at an equilibrium crossing. The hard-wall branch is declared already prepared, so its exterior product-state projection is not applied a second time.

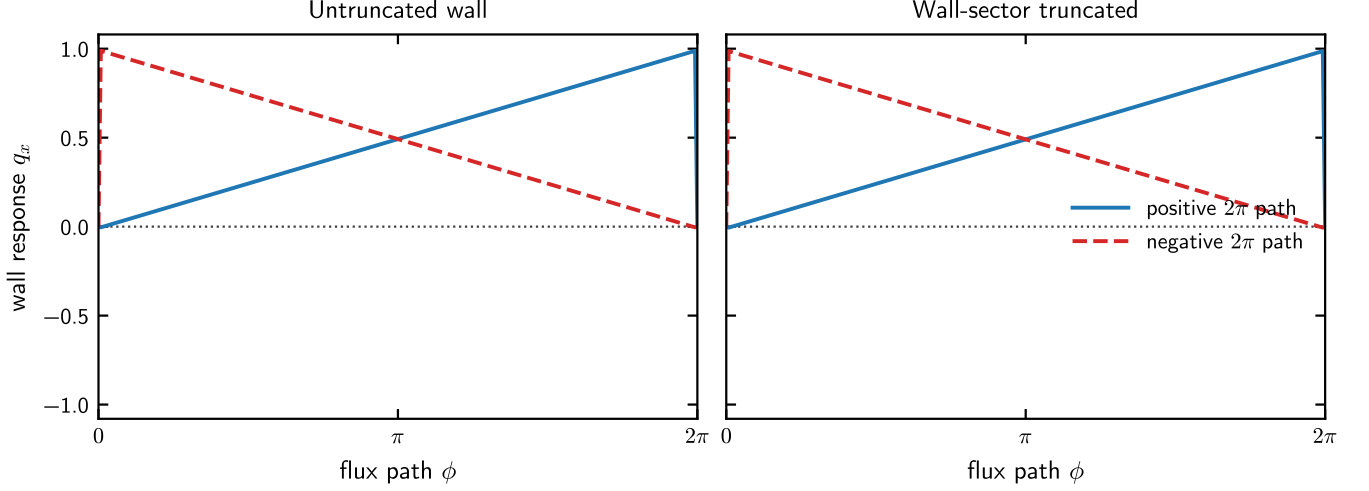


FIG. 1. Bare instantaneous equilibrium response at 20×40 . Each flux point is filled independently at fixed particle number. The response can be order one near the wall crossing, but the endpoint closes after a full flux quantum.

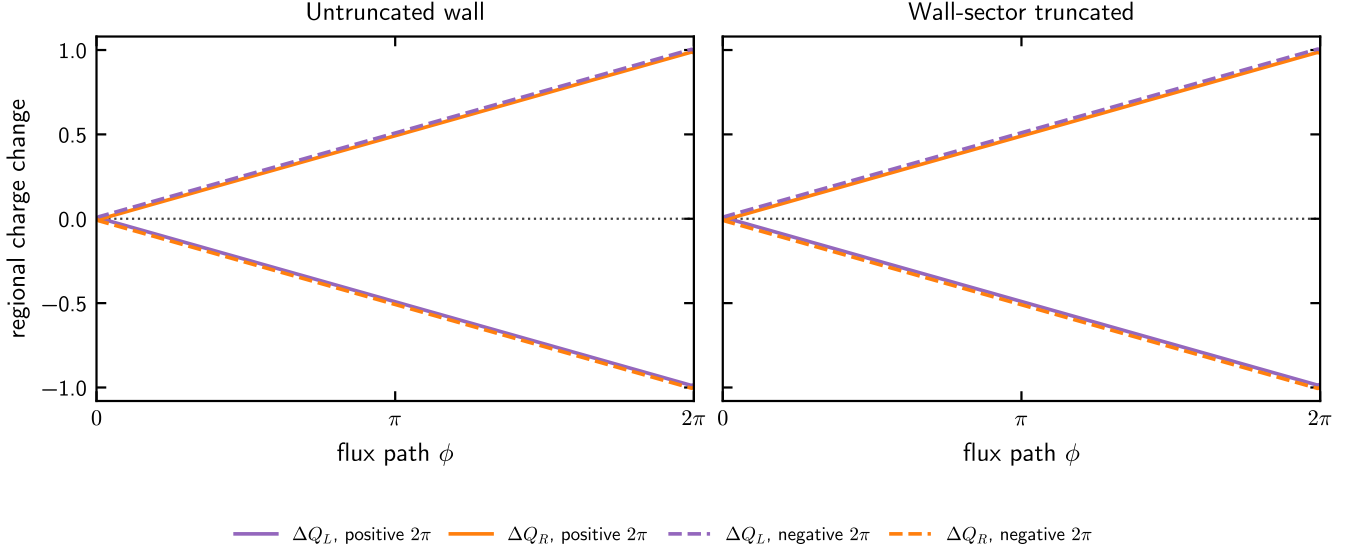


FIG. 2. Bare continued equilibrium regional charges. Overlap continuation carries the occupied wall branch through the crossing, leaving nearly one particle transferred between the two interfaces after one flux quantum. Left and right changes are equal and opposite.

The task table contains 20 burn-ins and 40 ramps. The two directions for a given (wall, sample ID) bind to the same burn-in file checksum, while their continuation seeds are distinct. Every physical evolution calls the canonical `classA.U1FGTN.run_markov_circuit` entry point; the campaign runner does not reproduce the Markov loop.

B. Raw and source-corrected observables

Online perfect correction can change particle number. For event e , let $m_e \in \{0, 1\}$ be the measured OW occupa-

tion, $s_e \in \{0, 1\}$ its correction target, and

$$\delta n_e = s_e - m_e. \quad (7)$$

For the normalized OW mode $|\chi_e\rangle$, the cumulative regional feedback source through ramp cycle j is

$$A_a(j) = \sum_{e \leq j} \delta n_e \langle \chi_e | R_a | \chi_e \rangle, \quad a = L, R. \quad (8)$$

The raw changes are referenced to the common branch origin, $\Delta N_a(j) = N_a(j) - N_a(0)$. We retain both the requested raw response and the redistribution after re-

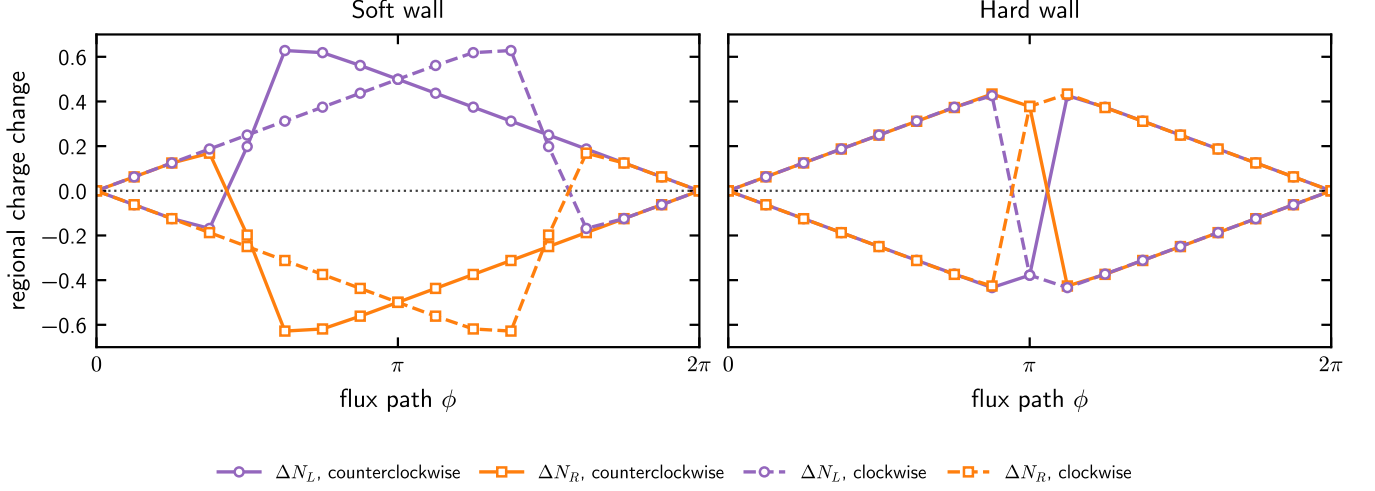


FIG. 3. Direct regional response of the two completed frozen-record circuit trajectories. Solid/circular and dashed/square curves denote increasing and decreasing flux. Charge moves oppositely between the two half-system regions and returns at 2π .

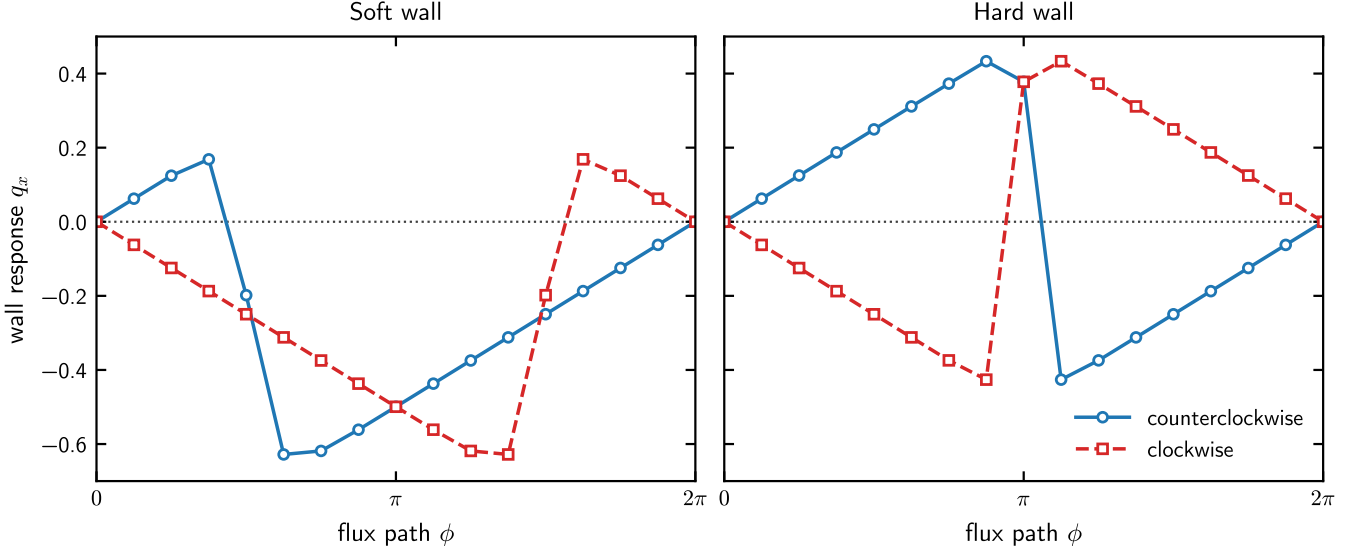


FIG. 4. Bare frozen-record wall response q_x for the soft and hard monitored circuits. Both individual records display an order-one, direction-sensitive response, but the closed endpoint distinguishes this conditional static family from the continued equilibrium pump.

moving the directly assigned feedback source,

$$q_a(j) = \Delta N_a(j) - A_a(j), \quad (9)$$

$$q_x^{\text{corr}}(j) = \frac{q_R(j) - q_L(j)}{2}. \quad (10)$$

The event weights must partition unity. At every cycle boundary the runner checks

$$\Delta N_{\text{tot}}(j) = \sum_{e \leq j} \delta n_e, \quad (11)$$

$$q_L(j) + q_R(j) = 0 \quad (12)$$

to absolute tolerance 10^{-9} . Compact products include all 17 values of ϕ , raw and corrected regional charges, raw and corrected q_x , total charge, occupied-frame rank, net injection, feedback-event count, continuity residuals, and the x -resolved density and source profile. No covariance history is stored.

C. Ensemble analysis and claim boundary

The independent resampling unit is the wall-specific burn-in sample ID. The clockwise and counterclockwise descendants of an ID remain paired in every bootstrap draw.

The preregistered analysis reports trajectory means and 95% trajectory-bootstrap intervals for $\Delta N_{L,R}$, q_x , their corrected counterparts, the direction-odd and direction-even combinations, endpoint distributions, and wall-resolved density changes. Direction parity is formed at common ramp coordinate $|\phi|$:

$$q_x^{\text{odd/even}}(|\phi|) = \frac{q_x^{(+)}(|\phi|) \mp q_x^{(-)}(|\phi|)}{2}, \quad (13)$$

with the upper sign giving the odd component.

Quantization is not an acceptance criterion. This is a short, rapidly driven monitored trajectory with stochastic feedback, not quasistatic continuation of an equilibrium occupied projector. The primary questions are whether the ensemble-averaged response is direction odd, wall localized, reproducible in both wall constructions, and distinct from its direct feedback source.

VI. COMPLETED CONTINUOUS FROZEN-RECORD RAMP

The next calculation isolates flux response from Born sampling noise. It reuses, without modification, the twenty verified cycle-40 occupied frames of the online campaign. For each wall-specific sample, one further 64-cycle zero-flux trajectory is generated from that frame and its complete raster- y measurement record $\mathbf{m}$ is frozen. The same starting frame and the same prefix of $\mathbf{m}$ are then used for the two directional ramps. Consequently, the clockwise and counterclockwise descendants have identical measurement outcomes, correction targets, rank histories, and total injected charge. Only their OW functions differ.

For $M \in \{16, 32, 64\}$ and $\sigma = \pm 1$, the ramp is one continuous conditioned trajectory with

$$\phi_j^{(\sigma)} = \sigma \frac{2\pi j}{M}, \quad j = 0, \dots, M. \quad (14)$$

The frame at step j is the input to step $j+1$; the calculation never restarts from the burn-in frame at an intermediate flux. The first M cycles of the common 64-cycle record are used at ramp length M . There is no spectral offset because Eq. (14) describes physical state propagation rather than static eigenstate branch selection.

At all $M+1$ boundaries we save ΔN_L , ΔN_R , ΔN_{tot} , q_x , occupied-frame rank, injection count, x -resolved density, selected-record probability, and numerical continuity residuals. No source-subtracted response and no tangent spectrum enter this experiment. The primary endpoint test is whether

$$q_x^{(+)}(2\pi) \simeq -q_x^{(-)}(-2\pi) \quad (15)$$

and whether that result stabilizes as M increases. Integer transfer is not an acceptance gate. The independent sampling unit is the frozen record associated with one burn-in parent; uncertainty bands are one sample standard deviation across the ten parents.

The campaign contains 20 reference-record tasks and 120 replay tasks. Each task writes one atomic NPZ and then a checksum-binding completion JSON. Restarting verifies those pairs and repeats only missing or invalid tasks. A sample-0 pass over both walls and all three ramp lengths precedes the remaining ensemble, but contributes directly to the same production output rather than forming a separate scientific sample. The versioned output root is `results/N16x20_frozen_record_continuous_ramp_s10_v1`, with root seed 2026090302. The launch uses 28 one-thread workers on physical cores 28–55, leaving the modular-handedness campaign on cores 0–27 untouched.

A. Continuous-ramp results

All 20 reference records and all 120 ramp products passed checksum and pairing verification. The largest charge-continuity residual was 1.08×10^{-11} . Define the paired direction-odd endpoint by

$$q_{x,\text{odd}}(M) = \frac{q_x^{(+)}(2\pi) - q_x^{(-)}(-2\pi)}{2}. \quad (16)$$

For the soft wall its mean and sample standard deviation at $M = (16, 32, 64)$ are $(0.7122 \pm 0.2234, 0.4959 \pm 0.1513, 0.1475 \pm 0.1655)$. The hard-wall values are $(0.9139 \pm 0.0932, 0.7415 \pm 0.1930, 0.3199 \pm 0.1927)$. The signal is handed and reproducible at the faster ramps, but decreases as the ramp is slowed. In this accessible range it therefore does not approach a nonzero quantized adiabatic pump.

VII. PREREGISTERED FIXED-FLUX QUENCH

The next calculation asks a distinct nonadiabatic question. For each of the same 20 burn-in frames and 64-cycle records, the frame is reset and every OW projector in the replay is held at

$$\phi = \sigma\varphi, \quad \varphi \in \left\{ \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi - 10^{-7} \right\}, \quad \sigma = \pm 1. \quad (17)$$

The constant schedule is installed before the cycle-zero observation and remains unchanged for all 64 cycles. This is a sudden change of the measurement channel, not a ramp.

Let $N_a^\phi(t)$ be the half-system charge obtained from the fixed-twist replay and $N_a^0(t)$ the charge in the zero-flux generation of the same record. The direct paired response is

$$\delta_\phi N_a(t) = N_a^\phi(t) - N_a^0(t), \quad (18)$$

$$q_x^\phi(t) = \frac{\delta_\phi N_R(t) - \delta_\phi N_L(t)}{2}. \quad (19)$$

Equation (19) is a same-record state difference; it is not a subtraction of an analytically assigned feedback source.

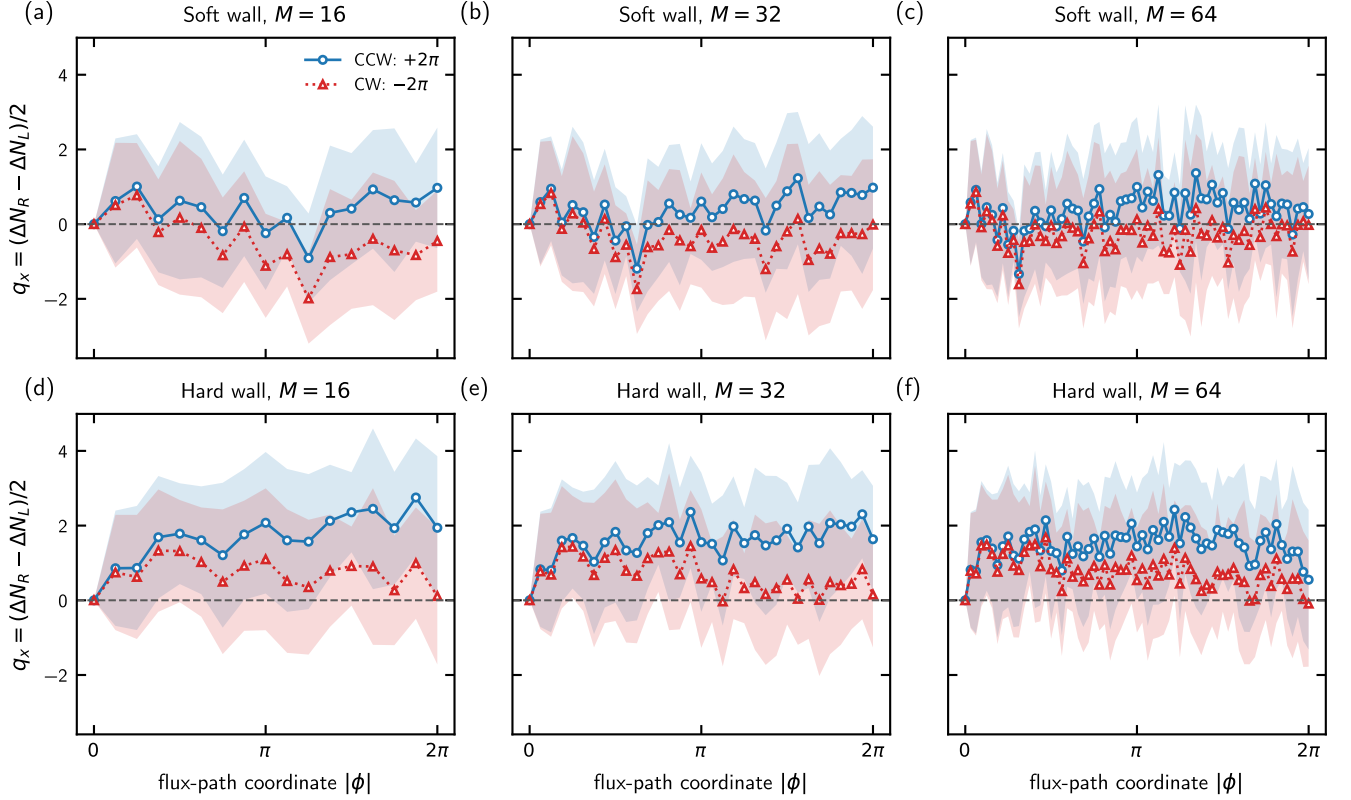


FIG. 5. Direct continuously replayed frozen-record response for ten paired records per wall. Curves show trajectory means and shading denotes one sample standard deviation. The same ordered outcomes are imposed in the two directions at each M .

Clockwise and counterclockwise partners have identical records, rank histories, and feedback-injection histories. The paired odd response $[q_x^{+\varphi}(t) - q_x^{-\varphi}(t)]/2$ is primary, while the even component diagnoses record-specific redistribution.

The near- 2π point tests large-gauge closure after a sudden projector quench. A small response would establish that the fixed channel remains periodic; a transient response would quantify the failure of the untwisted burn-in state to be stationary under the gauge-shifted conditioned channel. Neither outcome is interpreted as an adiabatic or quantized pump. The campaign contains 160 atomic replay tasks, uses root seed 2026090303, and writes to `results/N16x20_frozen_record_fixed_flux_quench.s10.v1`.

VIII. COMPLETED FINITE-DIFFERENCE PARENT-HAMILTONIAN RAMP

A. State-derived Hamiltonian and physical evolution

The static projector continuation above answers which occupied subspace is obtained by following eigenmodes through a flux crossing; it does not solve a time-dependent Schrödinger equation. To make that distinction opera-

tional, we reuse 50 verified endpoints from the completed $N_x \times N_y = 20 \times 24$ monitored ensemble: sample IDs $0, 4, \dots, 96$ for each of the soft and hard walls. This every-fourth-ID selection was fixed independently of the earlier zero/unit pump outcome.

For each endpoint occupied frame F_0 , define

$$P_0 = F_0 F_0^\dagger, \quad h_0 = \mathbf{1} - 2P_0. \quad (20)$$

The entire Hamiltonian path is constructed from this fixed h_0 ; it is not rebuilt from the evolved state. With minimum-image periodic displacement $d_y(r, s)$, the threaded parent is

$$h_{rs}(\phi) = (h_0)_{rs} \exp \left[i\phi \frac{d_y(r, s)}{N_y} \right], \quad (21)$$

followed by explicit Hermitian symmetrization at roundoff level. The flux is ramped in both orientations,

$$\phi_\sigma(t) = \sigma \frac{2\pi t}{T}, \quad \sigma = \pm 1, \quad T = 10^4, \quad (22)$$

and the occupied frame obeys

$$i \frac{dF}{dt} = h[\phi_\sigma(t)] F. \quad (23)$$

This is deterministic unitary evolution under a constructed one-particle parent Hamiltonian. It is neither

a rerun of the monitored circuit nor an instantaneous spectral reprojection.

B. Finite-difference scheme and observables

Equation (23) is integrated in complex128 using classical explicit fourth-order Runge–Kutta with the Hamiltonian evaluated at the beginning, midpoint, and end of every step. The loop contains 128 equal-flux observation intervals and 320 RK4 steps per interval, hence 40960 steps and

$$\Delta t = \frac{10^4}{40960} = 0.244140625. \quad (24)$$

At observation boundaries a thin QR factorization removes accumulated occupied-frame gauge and norm error. Right-unitary changes of F leave $FF^\dagger$ invariant, so this stabilization is not an energetic reprojection. Maximum Gram residuals are retained before and after every QR.

At all 129 observation points we save ϕ , t , ΔN_L , ΔN_R , q_x , total-charge residual, x -resolved density, instantaneous left current, and parent-Hamiltonian expectation value. Endpoint diagnostics include overlap with the large-gauge-transformed source frame, overlap with the independently filled instantaneous occupied subspace, its excitation number, and the instantaneous occupied–empty gap. Each of the 100 wall/direction paths writes one compact result/completion pair. Because a single path is long, one rolling occupied-frame checkpoint is updated every four flux intervals; a restart repeats at most that interval block.

The production discretization is checked by an independent four-path step-halving calculation using sample 0 for both wall constructions and both orientations. That calculation uses 640 RK4 steps per observation interval, $\Delta t = 0.1220703125$, and a separate configuration and output root. We compare both the endpoint and the maximum pointwise difference in q_x ; the four paths are a numerical convergence test and are not included as additional statistical samples.

C. Finite-size crossing prediction and observed response

At finite circumference the wall crossing is generally avoided. The strictly slow limit can therefore follow the instantaneous occupied subspace through the avoided crossing and return to the original charge distribution, whereas diabatic passage can remain on the spectral-flow branch and leave an order-one q_x . In a local two-level approximation with minimum gap g and crossing slope s , the diabatic probability is

$$P_D \simeq \exp\left(-\frac{Tg^2}{4s}\right). \quad (25)$$

This formula was preregistered as a descriptive comparison, not an acceptance gate. Five-point fits to the already

saved gap curves give median $s = 0.74282$ for the selected soft states and 0.74296 for the hard states, with interquartile widths below 0.002. Before any RK4 path completed, Eq. (25) predicted $P_D > 1/2$ for 16 of 25 soft states and 13 of 25 hard states. The primary analysis tests the endpoint direction-odd q_x against this gap prediction and against each state’s prior static continued-projector classification. Neither integer transfer nor adiabatic closure was imposed as an acceptance criterion. All 100 paths are now checksum verified. The raw soft-wall endpoint means are $q_x = +0.71569$ for CCW and -0.71422 for CW; 18 of 25 paths in each direction have $|q_x| > 0.5$. For the hard wall the corresponding means are $+0.53622$ and -0.53624 , with 13 of 25 paths above 0.5 in each direction. Charge conservation is better than 2.6×10^{-13} and the maximum post-QR Gram residual is 2.7×10^{-14} . The endpoint response is strongly sample dependent and nearly antisymmetric under direction reversal.

To distinguish transport from final finite-size closure, the preregistered summary reports three quantities for each saved endpoint: the direction-odd q_x at $|\phi| = \pi$, the largest $|q_x|$ attained anywhere along the loop and its flux location, and the residual direction-odd q_x at $|\phi| = 2\pi$. Endpoint paths are described as closed when $|q_x| < 0.25$, order-one when $|q_x| > 0.75$, and intermediate otherwise; these labels summarize the distribution and are not pass/fail thresholds.

D. Canceled physical-time width repeat

The sample-dependent avoided-crossing interpretation was to be tested by increasing only the separation between the two walls. A second calculation reused 25 soft-wall and 25 hard-wall endpoints from the completed $N_x \times N_y = 24 \times 24$ monitored ensemble, again selecting IDs 0, 4, $\dots$, 96 without reference to their static pump outcomes. Its walls are at $x = 6, 18$, so both separations equal 12 sites, compared with 10 sites in the 20×24 calculation. The circumference, flux grid, ramp time, Runge–Kutta step, wall definitions, subsystem partition at $N_x/2$, and all analysis thresholds remain unchanged. Because the two geometries were generated with independent monitored-trajectory seeds, directions are paired within each endpoint but statistical comparisons between $N_x = 20$ and 24 are unpaired.

Before any 24×24 RK4 path was evaluated, its saved static continuation data gave 23 of 25 soft-wall events and 11 of 25 hard-wall events under $|q_x^{\text{odd}}(2\pi)| > 0.5$. The corresponding mean static responses are 0.91588 and 0.43988, while the median minimum occupied–empty gaps are 0.00898 and 0.01738. These values and the complete endpoint selection are stored in a checksum-bound preregistration record. The test asks whether the larger wall separation shifts the gap distribution, the direction-odd charge distribution, or the fractions above 0.5 and 0.75. None is an acceptance gate.

The physical-time 24×24 run was stopped on Septem-

ber 6, 2026, because the scientific question was replaced by geometric Kato continuation. At the stop, no path had completed, 28 rolling NPZ/JSON checkpoint pairs were verified, every checkpoint was at observation interval 104 of 128, and 72 paths had not started. The configuration, log, and all checkpoints remain in the original versioned output directory. The exact inventory is recorded in `cancellation_inventory.json`; the stopped artifacts are not interpreted as results and are not overwritten.

IX. PREREGISTERED ADAPTIVE KATO CONTINUATION

The replacement width experiment transports the overlap-selected spectral projector geometrically rather than evolving a state in physical time. It uses the same endpoint IDs at 20×24 and 24×24 , both walls, and both flux orientations. At every RK4 stage, eigenvectors of the twisted state-derived parent are ranked by their previous-projector weights; the fixed-rank selected projector supplies the analytic Kato generator $[\partial_\phi P_{\text{spec}}, P_{\text{spec}}]$. One full step is compared with two half steps and accepted only when their normalized projector discrepancy is at most 10^{-8} . The selected-excluded gap is saved, but does not alone choose the step.

The complete theory, derivation from $P^2 = P$, analytic projector derivative, adaptive controller, regulator convention, diagnostics, and resume protocol are fixed in the companion two-column methods note `kato_overlap_continuation_methods.tex`. The campaign contains 200 paths and reports raw CW and CCW q_x separately. Quantization, agreement with the 64- and 128-grid static classifications, and width dependence are pending measured outcomes, not acceptance gates.

X. COMPLETED WALL-DIABATIZED SPECTRAL FLOW OF MONITORED ENDPOINTS

The final spectral calculation uses the same direct regional charge as Eq. (3), but changes how the occupied subspace is continued at the rank crossing. The input matrix contains $N_y = 24$ and $N_x = 20, 24, 28, 32$, soft and hard walls, and both $n_{\text{shell}} = 1$ and the dense OW construction. Each of the 16 ($n_{\text{shell}}, N_x, \text{wall}$) cells contains $S = 100$ independent pure, half-filled monitored endpoint states prepared for 48 raster- y cycles with perfect correction and no postselection. No monitored dynamics is rerun during flux threading.

For every endpoint projector, the single-particle parent is threaded through 256 intervals in each direction. Ordinary occupied spectator states are continued by previous-projector overlap. The two modes at the occupied-empty boundary are removed from that global assignment, their isolated cluster is aligned between neighboring flux points, and $E^\dagger(R_R - R_L)E$ defines left- and right-localized wall modes. The wall label occupied on entering the cross-

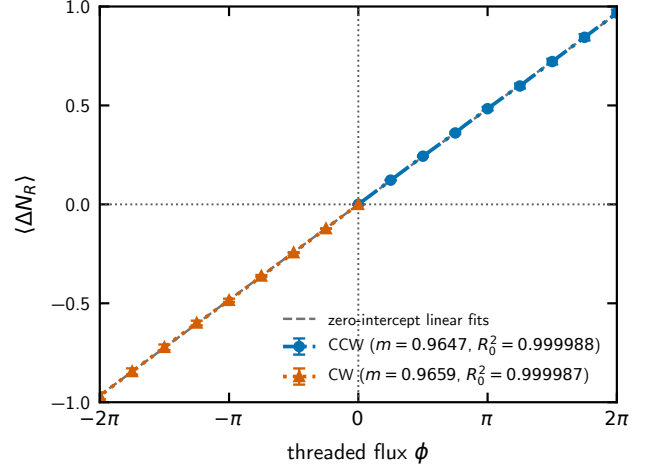


FIG. 6. Chiral right-subsystem charge response for the 32×24 hard wall with $n_{\text{shell}} = 1$. The curves show the raw trajectory mean $\langle \Delta N_R \rangle$ for $S = 100$ independent endpoint states per direction, and the vertical bars show $\pm \text{SEM}$. The signed axis displays CW on $-2\pi \leq \phi \leq 0$ and CCW on $0 \leq \phi \leq 2\pi$. Charge conservation gives $\Delta N_L = -\Delta N_R$ to numerical precision, so the redundant left-subsystem curve is omitted. Gray dashed segments are separate zero-intercept fits in normalized signed flux; the corresponding slopes and through-origin coefficients of determination are reported in the legend.

ing is retained through it and then recombined with the spectator subspace. A path is marked unresolved when the pair is not isolated or oppositely wall localized, but unresolved paths remain in every raw distribution and mean below.

Figures 7 and 8 show the complete flux-resolved response. CCW and CW are plotted separately; no q_x^{odd} combination is used. The curves expose an almost linear spectral transfer from zero to the orientation-expected unit response. The standard-deviation bands are controlled mainly by a small zero-transfer subpopulation rather than by broadly fractional sample-wise responses.

Figure 6 isolates the requested chirality diagnostic: the raw right-subsystem charge at $N_x \times N_y = 32 \times 24$ for the hard wall and $n_{\text{shell}} = 1$ only. CW is placed on the negative signed-flux axis and CCW on the positive signed-flux axis. The vertical error bars are the standard error of the mean, $s/\sqrt{100}$, rather than a confidence interval. Separate zero-intercept least-squares fits use $\langle \Delta N_R \rangle = m\phi/(2\pi)$ and give $m = 0.9647$, $R_0^2 = 0.999988$ for CCW and $m = 0.9659$, $R_0^2 = 0.999987$ for CW.

Of the 3200 directional paths, 3145 (98.3%) have the correct orientation and $|q_x| > 0.9$; 3120 (97.5%) satisfy $0.95 \leq |q_x| \leq 1.05$. Forty-eight paths (1.5%) lie within 0.1 of zero, and only seven lie between the near-zero and near-unit sectors. The broadest cell is the $n_{\text{shell}} = 1$ hard wall at $N_x = 28$, with raw endpoint means $+0.9177$ for CCW and -0.9174 for CW. The dense soft-wall cells at $N_x = 28, 32$ give unit response for all 200 directional paths

$n_{\text{shell}} = 1$: monitored-endpoint wall spectral flow

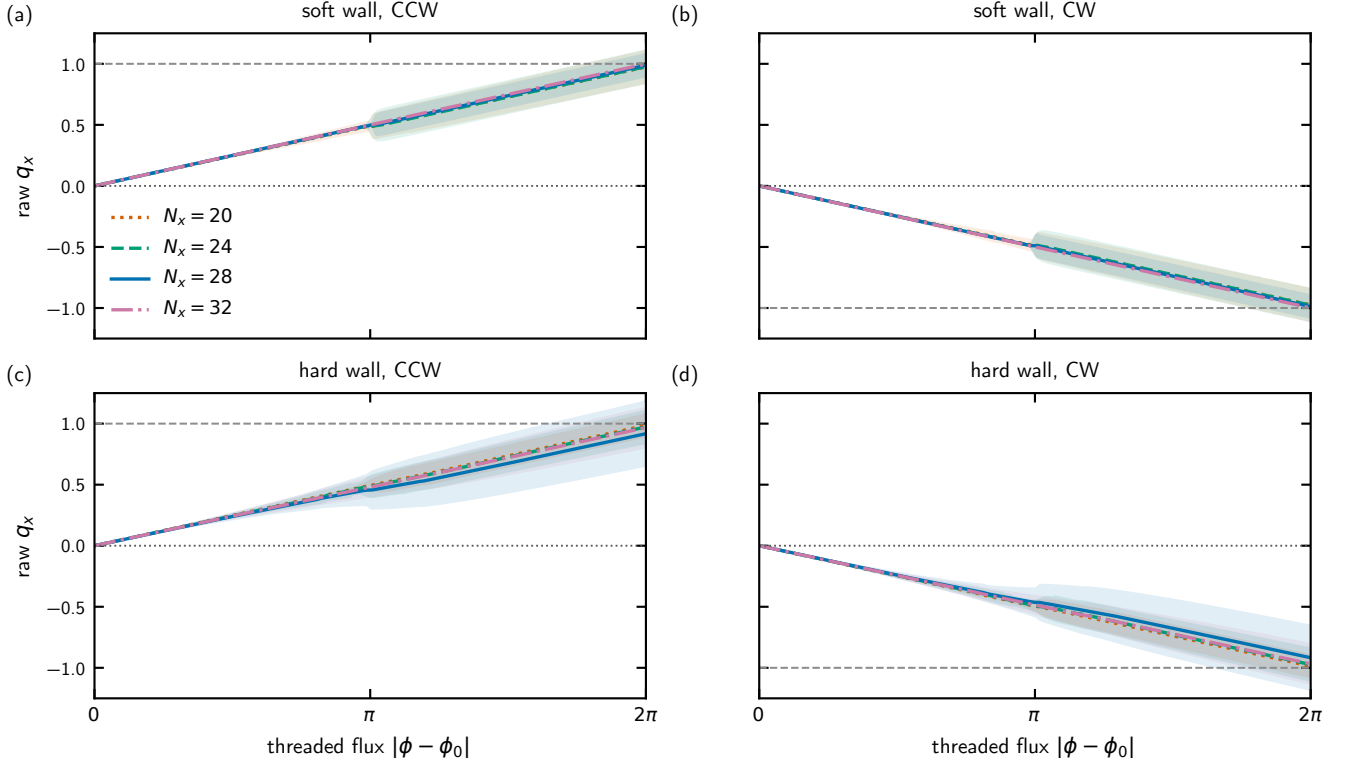


FIG. 7. Raw flux-resolved wall response for $n_{\text{shell}} = 1$. Rows show soft and hard walls and columns show CCW and CW paths. Each curve is the mean of $S = 100$ independent monitored endpoint states at fixed N_x ; shaded bands are one trajectory standard deviation. Colors and line styles distinguish $N_x = 20, 24, 28, 32$ at fixed $N_y = 24$. Every source trajectory starts from a pure half-filled state and runs for 48 raster- y cycles. The estimator is the 256-interval wall-diabatized spectral continuation, and unresolved paths are retained.

at each width to the displayed precision. The wall cluster is qualified for 1569 of 1600 endpoints; the 31 unresolved endpoints are retained in these numbers.

The maximum total-charge residual over the reconciled matrix is 4.55×10^{-13} , the maximum frame/projector residual is 9.59×10^{-14} , and the largest forward-reverse undo error is 2.33×10^{-15} . Thus the small non-unit population is not numerical charge leakage. The improvement over the earlier binary ordinary-overlap calculation comes from specifying the wall branch inside an independently qualified edge cluster, rather than allowing the global rank selector to make a grid-sensitive choice at the finite-size avoided crossing.

The executable companion `notebooks/equilibrium_wall_crossing_explorer/equilibrium_wall_crossing_explorer.ipynb` uses a deterministic translation-invariant 20×24 equilibrium OW slab to expose this crossing with two flux sliders. It displays the energy splitting, exchange of left/right character, wall-localized basis, instantaneous and continued occupations, and direct q_x . Its saved default run gives $q_x(2\pi) = 0.9999999992$ for overlap continuation while

instantaneous refilling closes within 3.5×10^{-10} .

XI. EXECUTION, PROVENANCE, AND STATUS

Each fixed-flux replay writes one compressed NPZ followed by one small completion JSON. The completion record binds task ID, wall, sample ID, fixed twist, seed, scientific configuration hash, source hashes, result byte count, and SHA-256, as well as the exact burn-in and reference-record SHA-256 values. Restart verifies both members and reruns only a missing or invalid task. There is no intra-task checkpoint, server transaction layer, migration ledger, or background Drive telemetry.

The fixed-flux production root is `results/N16x20_frozen_record.fixed_flux_quench.s10.v1`. The root seed is 2026090303. The host launch uses 28 single-thread workers on physical cores 28–55 and leaves the modular-handedness calculation on cores 0–27 untouched. Progress bars report completed replay cycles and tasks. Analysis begins only after all 160 replay

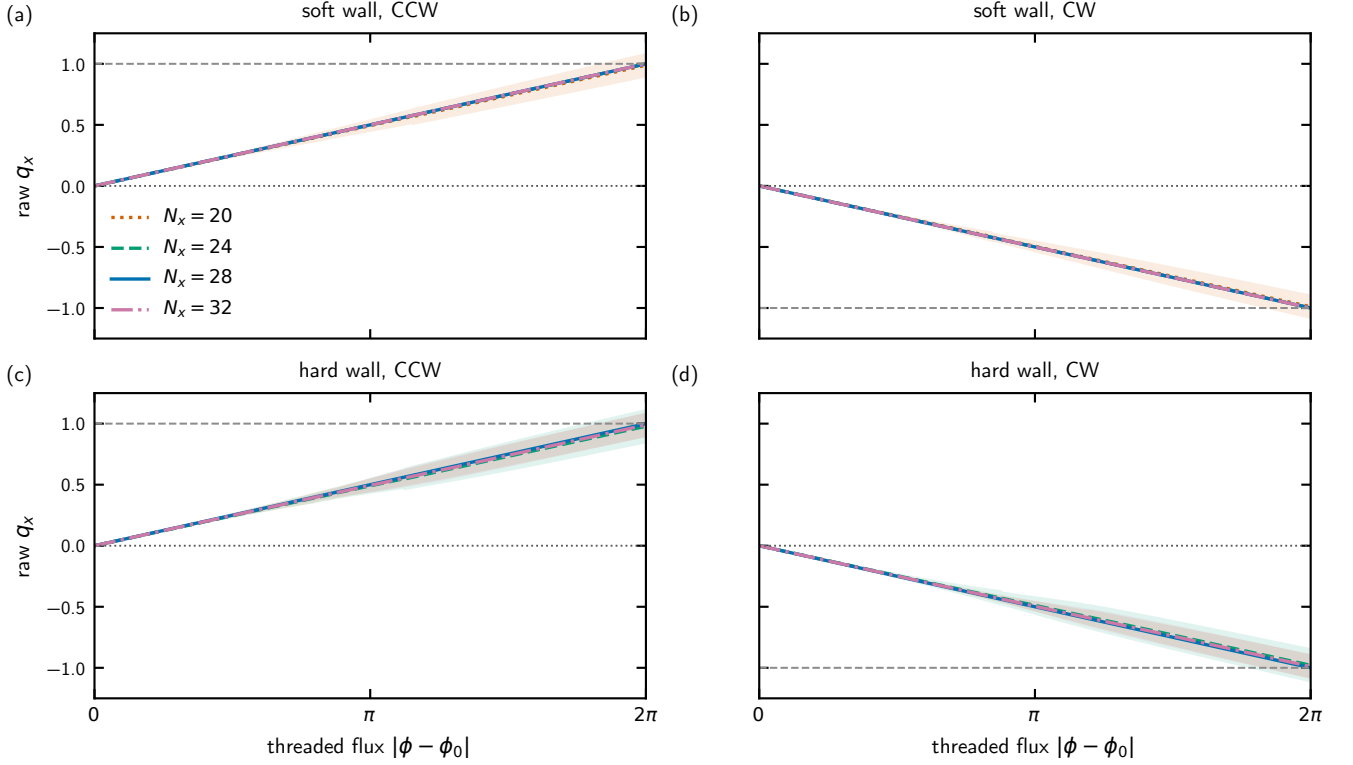
dense n_{shell} : monitored-endpoint wall spectral flow

FIG. 8. Raw flux-resolved wall response for the dense OW construction, with the same geometry, initialization, 48-cycle preparation, 256-interval estimator, $S = 100$ independent sampling unit, and trajectory-standard-deviation uncertainty as Fig. 7. Dense soft-wall curves at $N_x = 28, 32$ are nearly coincident and reach unit response with negligible inter-trajectory spread.

products verify.

TABLE I. Campaign status when the fixed-flux quench was preregistered. “Pending” is a statement about saved products, not a physical result.

calculation	independent samples
exact flattened Hamiltonian	deterministic
static frozen record	1 soft + 1 hard
online ramp burn-in	10 soft + 10 hard
online $\pm 2\pi$ ramps	20 per direction
continuous frozen records	10 soft + 10 hard
continuous frozen ramps	60 per direction
fixed-flux quenches	80 per direction
20×24 physical-time RK4	50 states, 2 directions
24×24 physical-time RK4	50 states, 2 directions
adaptive Kato width test	100 states, 2 directions
wall-diabatized spectral width sweep	1600 states, 2 directions

No statement about the sign, magnitude, quantization, or width dependence of the adaptive Kato result is made before those products exist. All earlier results and the

canceled physical-time checkpoints are preserved in their original versioned directories and are not overwritten by this campaign.

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